

Date: 11.08.2026

Domain: RL-Based Autonomous Intersection Navigation for Advanced Driver Assistance Systems (ADAS)

1. Project Type: Simulation-based Reinforcement Learning

Simulation Platform: MetaDrive

Primary Focus: Dynamic intersection navigation, safety, decision-making, collision avoidance

2. Objectives

The main objective of this project is to develop a Reinforcement Learning-based autonomous navigation system capable of making safe and intelligent driving decisions at road intersections.

Specific objectives

1. Develop a simulation-based ADAS environment for autonomous vehicle research.
2. Model dynamic and interactive intersection scenarios involving multiple road users.
3. Integrate vehicle-state and simulated perception sensors such as camera and LiDAR.
4. Construct an observation space suitable for Reinforcement Learning.
5. Develop a reward function that encourages:
 - Collision avoidance
 - Lane adherence
6. Train an RL agent to make steering/throttle/braking decisions.
7. Evaluate the trained agent under randomized traffic and intersection scenarios.

8. Measure safety and driving performance using quantitative metrics such as:

- Collision rate
- Route completion
- Average speed
- Time to complete intersection

3. Current implementation boundary:

The development of the project is completed upto 4 stages:

Project



MetaDrive



Simulator Interface



MetaDrive Adapter



MetaDrive Environment Initialization

The sensor → RL → training → evaluation portion is the next development stage

4. Dataset

For this project, the INTERACTION Dataset is a particularly relevant reference dataset because the project focuses on interactive intersection navigation.

The dataset contains naturalistic trajectories of traffic participants in highly interactive driving scenarios. It includes scenarios such as:

- Signalized intersections
- Unsignalized intersections
- Roundabouts



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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

- Merging
- Lane changing

It also provides HD semantic maps and vehicle/pedestrian trajectory files.

This makes it useful for:

- Scenario analysis
- Traffic behavior modeling
- Reinforcement Learning research
- Decision-making
- Planning verification
- Driving scenario generation

The dataset's official website specifically lists reinforcement learning, decision-making/planning, and scenario generation among its intended applications.

Download / Access

Official dataset:

[INTERACTION Dataset — Official Website](#)

Dataset details and format:

[INTERACTION Dataset — Details & Format](#)

The architecture is:

Real-world INTERACTION Dataset



Scenario / Behavior Analysis



Scenario Design



MetaDrive



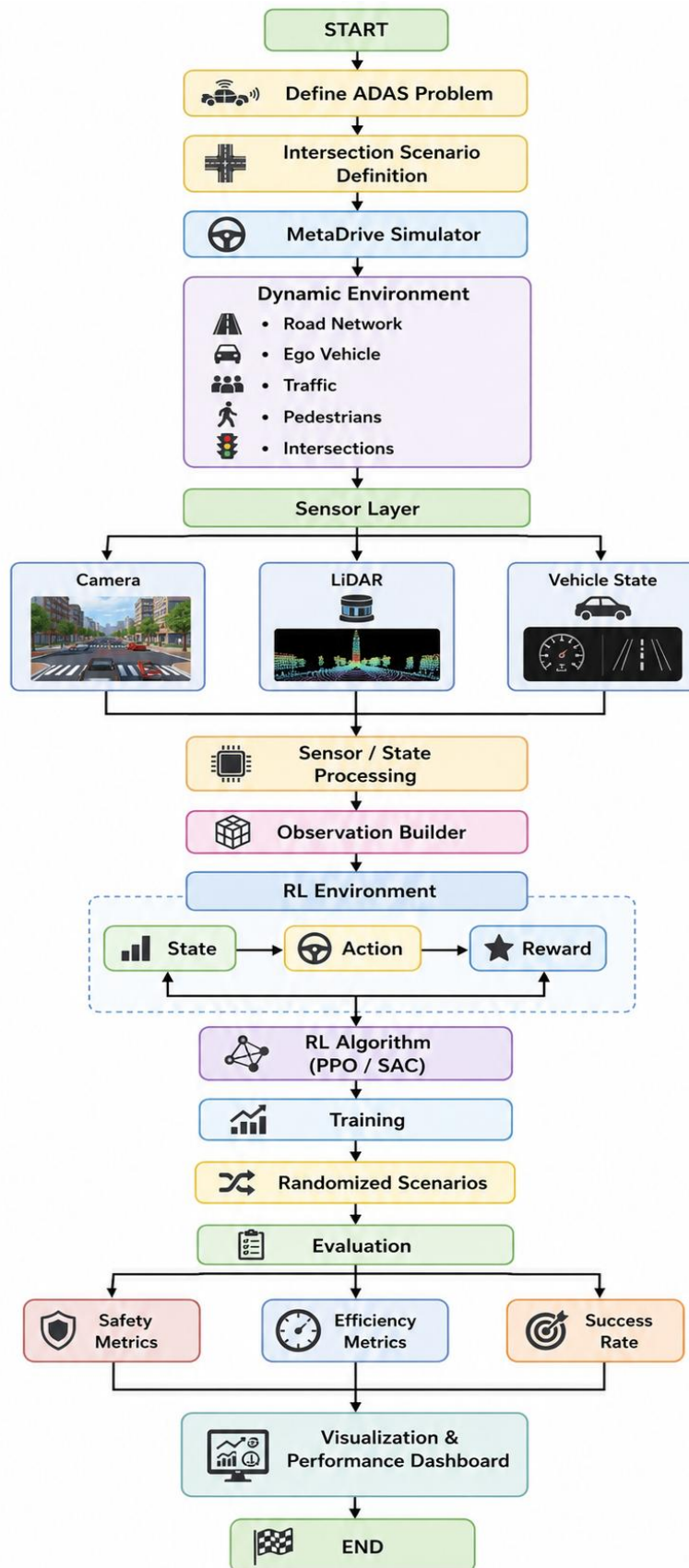
Synthetic Dynamic Training



RL Agent

This is a strong approach because we can use real-world trajectory information to make the simulated scenarios more realistic, while MetaDrive provides the environment in which the RL agent can actually interact and learn.

5.Workflow of the Project:



6. Methodology Overview

The complete methodology can therefore be summarized as:

INTERACTION Dataset

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Scenario / Behavior Analysis

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MetaDrive Simulation

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Dynamic Traffic Scenarios

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Sensor + Vehicle-State Data

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Observation Processing

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RL Environment

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Training

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Randomized Evaluation

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Safety + Performance Metrics

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ADAS Intersection Navigation Policy

Current Progress: The project foundation, Python environments, simulator abstraction, MetaDrive integration, and simulator initialization have been completed. Sensor integration, RL environment construction, agent training, dynamic scenario generation, and quantitative evaluation are the subsequent development stages.

Team Members

	Name	Roll Number
1.	N.Karthik	2420030511
2.	V.Sanjay	2420030385
3.	Ch.Srikar Reddy	2420030509